Special products

1

a False
$8^2 + 96 + 6^2$

b False
$y^2 - 14y + 49$

2

a $(x+3)^2 = x^2 + 6x + 9$

b $(x-3)^2 = x^2 - 6x + 9$

c $x^2 + 64 - 16x = (x-8)^2$

d $(x+6)^2 = x^2 + 12x + 36$

e $\left(8x^2 - 9\right)^2 = 64x^4 - 144x^2 + 81$

f $(-9x+2)^2 = 81x^2 - 36x + 4$

g $(2y+2)(2y-2) = 4y^2 - 4$

h $(3x-8)^2 = 9x^2 - 48x + 64$

3

a $t^2 + 10t + 25$

b $t^2 + 10t + 25$

c

i No **ii** Yes **iii** No **iv** Yes

4

a $x^2 + 6x + 9$

b $m^2 - 10m + 25$

c $s^2 + 8st + 16t^2$

d $x^4 + 6x^2 + 9$

e $81x^2 + 36x + 4$

f $16 - 40m + 25m^2$

g $16x^2 + 56xy + 49y^2$

h $27x^2 + 144xy + 192y^2$

i $81x^2 + 432xy + 576y^2$

j $147x^3 - 210x^2y + 75xy^2$

k $\dfrac{64x^2 - 144x + 81}{81}$

l $36y^2 + 6y + \dfrac{1}{4}$

m $\dfrac{n^2}{16} - 2 + \dfrac{16}{n^2}$

5

a $m^2 - 9$

b $x^2 - 49$

c $9x^2 - 64$

d $16x^2 - 9y^2$

e $16y^2 - 81$

f $36 - p^2$

g $49y^2 - 64$

h $121 - 81p^2$

i $100 - m^2$

j $384x^2 - 486y^2$

k $20x^3 - 405xy^2$

l $0.64x^2 - 81$

m $81x^2 - 0.16$

n $0.04x^2 - 121$

o $v^2 - \dfrac{1}{49}$

p $36x^2 - \dfrac{1}{9}$

q $36x^2 - \dfrac{64}{9}$

r $x^2 - \dfrac{4}{x^2}$

s 111

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6 Example answer:

$$\begin{aligned} 403^2 &= (400+3)^2 \\ &= 400^2 + 2 \cdot 400 \cdot 3 + 3^2 \\ &= 160\,000 + 2400 + 9 \\ &= 162\,409 \end{aligned}$$

7 Example answer:

$$\begin{aligned} 896^2 &= (900-4)^2 \\ &= 900^2 - 2 \cdot 900 \cdot 4 + 4^2 \\ &= 810\,000 - 7200 + 16 \\ &= 802\,816 \end{aligned}$$

8 Example answer:

$$\begin{aligned} 201 \cdot 199 &= (200+1)(200-1) \\ &= 200^2 - 1^2 \\ &= 40\,000 - 1 \\ &= 39\,999 \end{aligned}$$

9

a $16y^2 + 24y + 9 \text{ units}^2$

b $16y^2 + 24y + 9 \text{ units}^2$

10

a $62x^2 - 25y^2$

b $62x^2 - 25y^2$

11

a $6x^2 + 36x + 54 \text{ units}^2$

b $x^3 + 9x^2 + 27x + 27 \text{ units}^3$

12 $m^3 - m$

13

a $y^4 - 81$

b $y^5 - 81y \text{ km}$

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Additional questions

14

a $(a+b)^2 = (a+b)(a+b)$
$= a(a+b) + b(a+b)$
$= a^2 + ab + ab + b^2$
$= a^2 + 2ab + b^2$

b $(a-b)^2 = (a-b)(a-b)$
$= a(a-b) - b(a-b)$
$= a^2 - ab - ba + b^2$
$= a^2 - 2ab + b^2$

c $(a+b)(a-b) = a(a-b) + b(a-b)$
$= a^2 - ab + ba - b^2$
$= a^2 - b^2$

15

a $(20+1)^2 = (20^2 + 2(20) + 1) = 441$

b $(20-1)^2 = (20^2 - 2(20) + 1) = 361$

c $(30+3)^2 = (30^2 + 2(90) + 9) = 1089$

d $(50-3)^2 = (50^2 - 2(150) + 9) = 2209$

16 $4x^2 - 8x + 4 \text{ cm}^2$

17 Rewrite $105^2 - 104^2$ as the difference of squares $(105+104)(105-104)$ and simplify each binomial individually to get $(209)(1) = 209$

18

a Variable choices will vary.
Let x represent the side length of the square and let y represent the width of the border. Then the area of the square will always be $(x+y)^2 = x^2 + 2xy + y^2$

b Let $x = (2m+1)$ and let $y = 3$ so that $(x+y)^2$ becomes $(2x+1+3)^2 = (2x+4)^2$. Then the total area is $4x^2 + 16x + 16$

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